

SARASWAT ACADEMY

CLASS TEST

Class: IX Subject: Science

Chapter: Cell – The Building Block of Life

Time: 30 Minutes Maximum Marks: 20

Section A – Multiple Choice Questions (5 × 1 = 5)

1. The basic structural and functional unit of life is:
(A) Tissue
(B) Organ
(C) Cell
(D) Organ system
2. Who first observed cells in cork?
(A) Charles Darwin
(B) Robert Hooke
(C) Rudolf Virchow
(D) Schleiden
3. Which organelle is known as the powerhouse of the cell?
(A) Golgi apparatus
(B) Ribosome
(C) Mitochondrion
(D) Lysosome
4. The movement of water through a selectively permeable membrane is called:

- (A) Diffusion
- (B) Respiration
- (C) Osmosis
- (D) Photosynthesis

5. Which plastid contains chlorophyll?

- (A) Chromoplast
- (B) Leucoplast
- (C) Chloroplast
- (D) Vacuole

Section B – Assertion & Reason ($1 \times 1 = 1$)

6. Assertion (A): Plant cells do not shrink in size in concentrated sugar solution.

Reason (R): The rigid cell wall maintains the shape of plant cells.

- (A) Both A and R are true and R is the correct explanation.
- (B) Both A and R are true but R is not the correct explanation.
- (C) A is true but R is false.
- (D) A is false but R is true.

Section C – Case Study ($4 \times 1 = 4$)

Riya placed one potato piece in plain water and another in concentrated salt solution. After one hour, the potato in plain water swelled while the other shrank.

- (a) What process caused this change?
- (b) Which potato piece gained water?
- (c) Which property of the cell membrane is responsible?
- (d) Why did the potato in salt solution shrink?

Section D – Subjective Questions (10 Marks)

7. Differentiate between prokaryotic and eukaryotic cells. (2 Marks)
8. Explain any two functions of the cell membrane. (2 Marks)
9. Name any four cell organelles and write one function of each. (4 Marks)
10. Differentiate between mitosis and meiosis. (2 Marks)

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